

Bharatiya Vidya Bhavan's  
**SARDAR PATEL INSTITUTE OF TECHNOLOGY**  
(An Autonomous Institute Affiliated to University of Mumbai)  
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058  
**MID SEMESTER EXAMINATION**

Sr.No. **8498**

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|            |         |                 |                      |          |            |
|------------|---------|-----------------|----------------------|----------|------------|
| AY:        | 2025-26 | Session:        | Odd MSE October 2025 | Date:    | 08-10-2025 |
| Sem:       | 1       | Slot:           | 10:00am - 11:00am    | C.Code:  | CS413      |
| Exam Type: | Written | Block:          | 401                  | Seat No: | 2022600032 |
|            |         | Course Section: | 1                    |          |            |



Name of the candidate. SHLOKA MARDIKAR

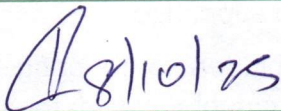
Candidate's UID number: 

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 0 | 2 | 2 | 6 | 0 | 0 | 0 | 3 | 2 |
|---|---|---|---|---|---|---|---|---|---|

Examination class room number: 

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|---|---|---|
| 4 | 0 | 1 |
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Invigilator's signature with date: 

|                                                                                   |
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(To be filled in by the candidate)

EXAMINATION: B.E. MSE B.Tech  
(For example FY MCA/FY BTECH)

SEMESTER: 7

PROGRAM: CSE-AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 8/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

| Question numbers        | 1 | 2 | 3 | 4 | 5 | 6 | Total | Signature of the examiner |
|-------------------------|---|---|---|---|---|---|-------|---------------------------|
| Marks awarded           |   |   |   |   |   |   |       |                           |
| Marks after revaluation |   |   |   |   |   |   |       |                           |

## INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



| Space For Marks | Question No. | START WRITING HERE<br>(Begin answer for each question on a new page)                                                                                                                                                                                                                                                                         |
|-----------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 | 1) a)        | $f(w) = \frac{1}{2} \ Xw - y\ ^2$ $f'(w) = \frac{1}{2} \times 2 \ Xw - y\  \cdot (X - 0)$ $= \ Xw - y\  X$ <p><del>XXX</del> Gradient descent.</p> $w = w_n = w_{old} - \eta \frac{\delta L}{\delta w}$ $x_n = x_0 - \eta \frac{\delta L}{\delta x}$ <p>Gradient descent is optimization algorithm used to improve the model's accuracy.</p> |



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1)b)

$$MSE = \frac{1}{N} \sum (y_i - \hat{y}_i)^2 \quad (\text{Mean squared error})$$

When bias is high and variance is low underfitting occurs. Bias <sup>makes</sup> is the model simple.

Overfitting occurs when the variance is high and bias is low. Variance high means there is a lot of randomness in the output. The model starts learning the noise of the data.

Instead of learning the underlying pattern, it learns the data completely. The model is too complex.

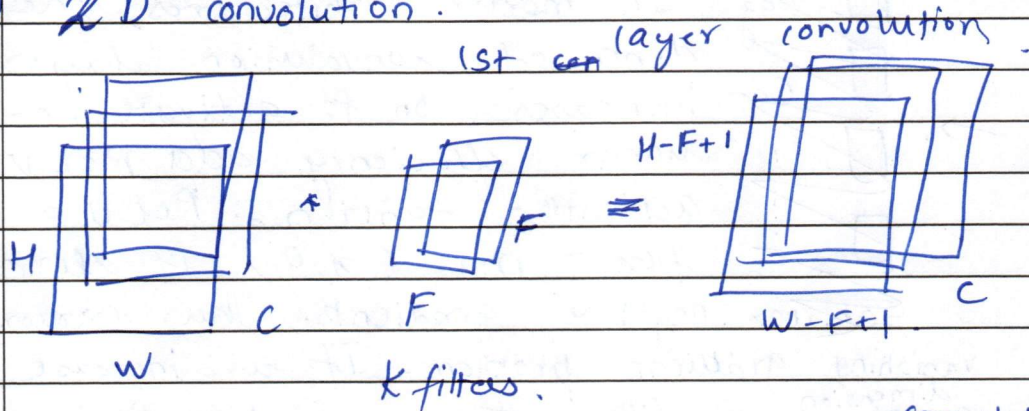
To avoid this we should use larger dataset, so model has more data to learn from. Apply regularization techniques L1, L2.

You can avoid it by early stopping i.e. pruning the tree after certain threshold. Fine-tuning, updating hyperparameters.

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2a) 2D convolution.

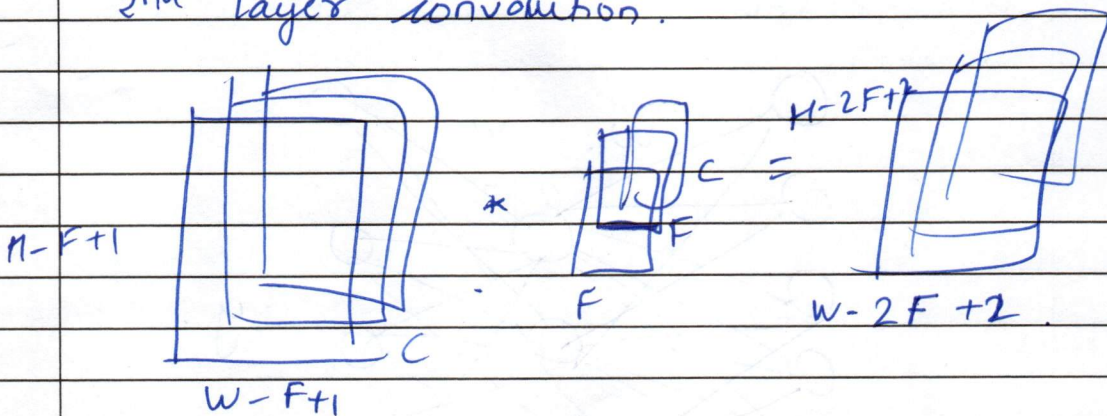


No. of parameters =  $\frac{\text{dimensions of filter} \times \text{bias}}{\text{No. of filters}}$

$$\text{No. of parameters} = (F \times F + 1) \times K$$

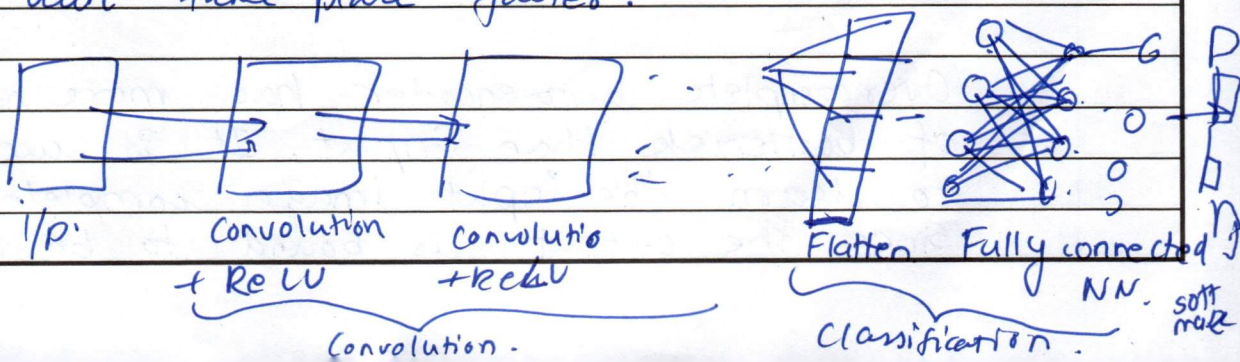
$$= (F^2 + 1) K$$

2nd layer convolution.



$$\text{No. of parameters} = (F \times F \times C + 1) \times K$$

depthwise separable convolutions undergo, a lot of computation of convolution from input image applying filters over and over again. The ~~input~~ size network efficiency might reduce, to avoid this apply pooling. It reduces the output size so convolutions can take place faster.



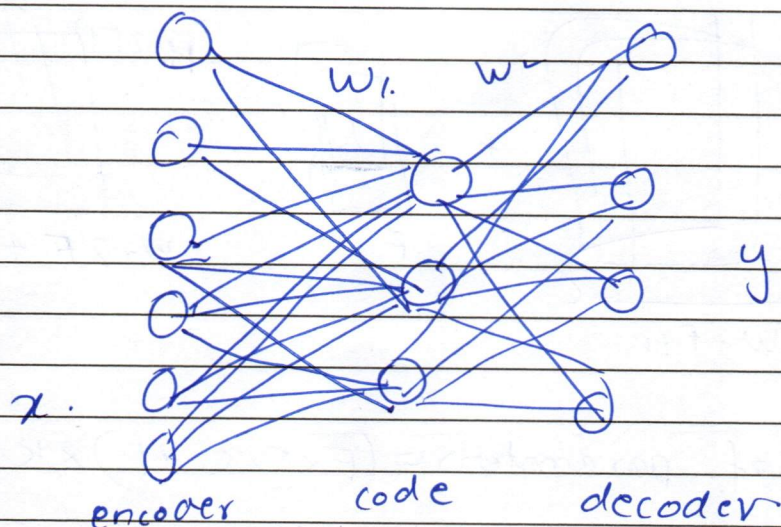
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☒ ~~Qog~~ In modern architectures, the depth of convolution layers increases, so to activate non-linear efficiency add ReLU activation function.  $ReLU = f(x) = \max(x, 0)$ . It eliminates ~~Softmax~~ negative gradients. Thus overcoming vanishing gradient problem. If we increase ~~dimension~~ <sup>dimension</sup> of filter, the parameter increases so does output image size. Complexity increases.

2b) Linear autoencoder has one layer of bottleneck, encoder and decoder. It is the simplest form of autoencoder.



$$y = (w_1, w_2) x$$

PCA :-

$$y = B^T x$$

Overcomplete autoencoder has more neurons of bottleneck than input. It is used to learn the input image completely. Since the output is bound to be a

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3/a) In RNN, loss function

$$L = -y_t \log(\hat{y}_t)$$

$$\frac{\delta L_3}{\delta w_3} = \frac{\delta L_3}{\delta y_3} \frac{\delta y_3}{\delta w_3} \quad \text{--- (1)}$$

$$y_3 = \text{softmax}(V_t h_t + b)$$

It depends on hidden state.

~~$$\frac{\delta y_3}{\delta w_3} = \delta$$~~

$$\therefore \frac{\delta L_3}{\delta w_3} = \frac{\delta L_3}{\delta y_3} \frac{\delta y_3}{\delta h_3} \frac{\delta h_3}{\delta w}$$

$$h_t = \tanh(V_t x_t + w_t h_{t-1} + b)$$

It depends on previous hidden states.

$$\frac{\delta L_3}{\delta w_3} = \frac{\delta L_3}{\delta y_3} \sum_{i=1}^3 \frac{\delta y_3}{\delta h_i} \frac{\delta h_i}{\delta w_3}$$

~~$$\frac{\delta L}{\delta h_{t-k}} = \frac{\delta L}{\delta h_{t-k}} \sum_i \frac{\delta h_{t-k}}{\delta h_i}$$~~

Vanishing gradient problem arises because ~~Backpro~~ Since tanh gradient ranges between 0 to 1. The multiplication will give smaller values. So, RNN will not be able to propagate back properly. LSTM uses gates to solve this problem. There are 3 gates. Forget gate, input gate, output gate. Forget gate:- selects which information to forget from previous hidden state.



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|                 |              | So, that contextual understanding increase.                                                             |
|                 |              | $f_t = \sigma(U_f x_t + W_f h_{t-1} + b_f)$                                                             |
|                 |              | Input gate $I_t$ uses sigmoid function ranges from 0 to 1. 0 to forget and 1 to retain the information. |
|                 |              | Input gate:- responsible for which information to send forward.                                         |
|                 |              | $i_t = \sigma(U_i x_t + W_i h_{t-1} + b_i)$                                                             |
|                 |              | Candidate gate:- stores the information, acts like memory cell.                                         |
|                 |              | $\hat{C}_t = \tanh(U_c x_t + W_c h_{t-1} + b_c)$                                                        |
|                 |              | cell gate:- $C_t = f_t \odot C_{t-1} + i_t \odot \hat{C}_t$                                             |
|                 |              | hidden state $h_t = o_t \odot \tanh(C_t)$                                                               |
|                 |              | $o_t = \sigma(U_o x_t + W_o h_{t-1} + b_o)$                                                             |

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3b) The gating equations of GRU are.

Reset gate:-

$$r_t = \sigma(U_r x_t + W_r h_{t-1} + b_r)$$

$r_t \rightarrow$  reset gate

$\sigma \rightarrow$  sigmoid function

$W_r \rightarrow$  weight ~~between~~ <sup>matrix of</sup> hidden to hidden layer.

$h_{t-1} \rightarrow$  previous hidden state

$b_r \rightarrow$  bias of reset gate

$U_r \rightarrow$  weight matrix of input and hidden layer.

Update gate:-

$$z_t = \sigma(U_z x_t + W_z h_{t-1} + b_z)$$

$\downarrow$  update gate       $\downarrow$  sigmoid       $\downarrow$  i/p.       $\downarrow$  previous hidden state

$W_z$  matrix of ~~between~~ hidden to hidden layer.

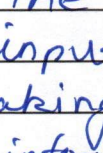
Candidate hidden gate state.

$$\hat{h}_t = \tanh(U_h x_t + W_h r_t h_{t-1} + b_h)$$

Hidden ~~gate~~ state.

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \hat{h}_t$$

Hidden state takes complement of update gate, that is responsible of ~~update~~ updating information in the cell state, which to take forward. The candidate hidden state, takes in the information that reset gate has decided to keep and multiplies it with update gate to decide the output.

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|-----------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |              | <p>LSTM has 3 gates :- Input gate, Forget gate and output gate.</p> <p>In GRU :- <del>It</del> The reset gate performs the tasks of input and forget gate of LSTM. Thus making it faster to commute. It takes the information from inputs decides which to discard and move forward.</p> $f_t = \sigma(W_f(h_{t-1}, x_t) + b_f)$ $f_t = \sigma(U_f x_t + W_f h_{t-1} + b_f)$ $i_t = \sigma(W_i x_t + W_i h_{t-1} + b_i)$ $\hat{C}_t = \tanh(W_c x_t + W_c h_{t-1} + b_c)$ <p>cell state:-</p> $C_t = f_t \odot C_{t-1} + i_t \odot \hat{C}_t$ $o_t = \sigma(W_o x_t + W_o h_{t-1} + b_o)$ <p>hidden state :-</p> $h_t = o_t \odot \tanh(C_t)$ <p>In LSTM we have to perform more operation due to higher no. of gates.</p> <p>GRU are used when we need the output faster pace. LSTM is computationally slow. GRU is used in language translation, text summarization.</p> |

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